Highly accurate protein structure prediction with AlphaFold

Paper by Jumper et al., *Nature* 596 (2021)

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`github.com/BenBullinger/ML4Genomics-Presentation-AF2`

Machine Learning for Genomics (263-5351-00L), ETH Zürich, Fall 2026

The problem: sequence to structure

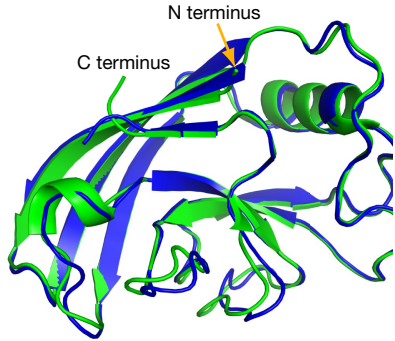
`SIFSYITESTGTPS...`

Primary amino acid sequence $\mathbf{s} \in \mathcal{A}^L$

Genomic sequencing yields $> 2 \times 10^8$ known proteins (UniProt); structural experiments cover $< 2 \times 10^5$ (Protein Data Bank, PDB).

Thermodynamic fold determination (Anfinsen) faces astronomical conformational search (10^{143} states, Levinthal).

Predict all heavy-atom 3D coordinates $\mathbf{x} \in \mathbb{R}^{N_{\text{atoms}} \times 3}$ directly from primary sequence.

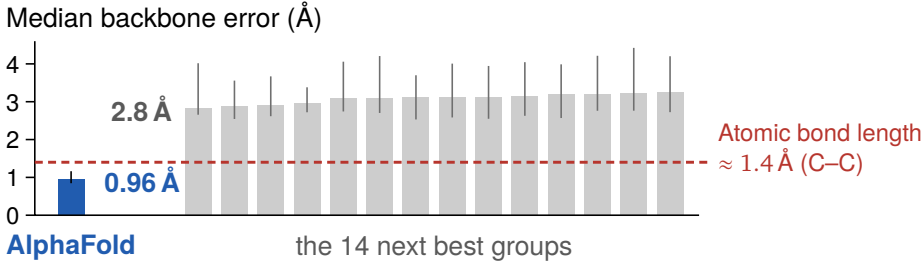


Ground truth in green (PDB 6Y4F), prediction in blue (0.8 Å RMSD)

[JEP+21] Jumper et al., *Nature* 2021, Fig. 1b. Target T1049 (PDB 6Y4F), CASP14.

The benchmark: CASP14

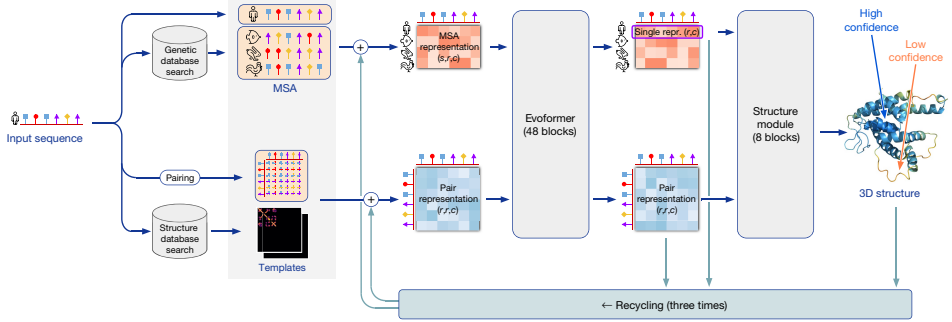
CASP14 blind assessment on unreleased structures: evaluated on Cα-r.m.s.d.$_{95}$ across 87 domains.



Median backbone error reaches 0.96 Å—surpassing typical covalent C–C bond length (1.4 Å).

Redrawn from Fig. 1a. Cα r.m.s.d.$_{95}$ on 87 domains (95% CIs). 1 Å = 10^{-10} m.

The pipeline: MSA in, coordinates out



Two-stage end-to-end pipeline: the Evoformer (48 blocks) refines sequence and pair tracks; the Structure Module (8 iterations) builds 3D coordinates.

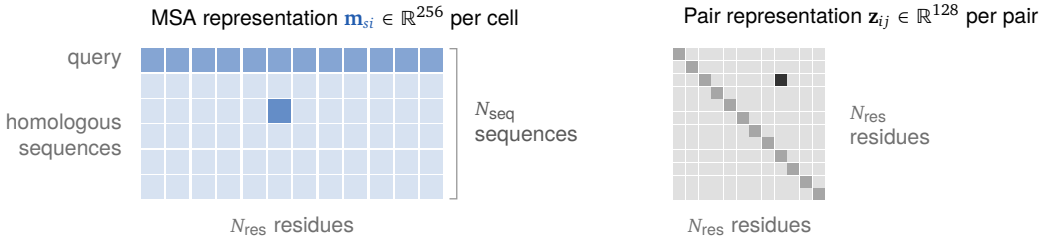
[JEP+21] Jumper et al., *Nature* 2021, Fig. 1e. Coordinates and pair features are recycled for 4 total passes.

Four facts, four design choices

		Fact	Design choice in AlphaFold 2
1		MSA is a set; chain is 1D	Axial attention on rows and columns
2		3D distances close triangles	Triangle updates on residue pair graph
3		Rigid motion invariance: SE(3)	Local frames; Invariant Point Attention
4		Proteins are chiral; mirrors differ	Frame Aligned Point Error (FAPE) loss

Removing any one of the four costs accuracy: the ablation study measures each.

The two representations

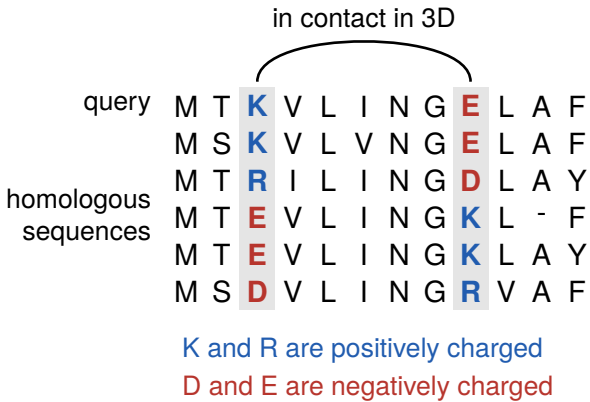


Evolutionary variation in $\mathbf{m}_{si}$; pairwise spatial geometry and orientation in $\mathbf{z}_{ij}$.

Dual-track representation: maintaining both tracks simultaneously avoids premature loss of raw evolutionary signals.

Supplementary Information (SI) 1.5, Algorithms 3 and 4. Sizes in training: 128 sequences, crops of 256 residues (SI Table 4).

The input: MSA representation



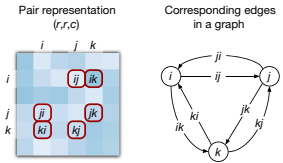
Compensatory mutations reflect spatial proximity: charge swaps preserve contacting salt bridges.

Raw alignments retain higher-order epistasis discarded by pairwise covariance tables (DCA).

End-to-end attention extracts contact geometry directly from evolutionary variation.

[MHC+11] D. S. Marks et al., *PLoS ONE* 6, 2011. The alignment is an illustration, not data.

The input: Pair representation



Edge features $\mathbf{z}_{ij} \in \mathbb{R}^{128}$ form a complete directed graph over all residue pairs.

Initialized from sequence separation $|i - j|$ (clipped one-hot) and homologous PDB templates.

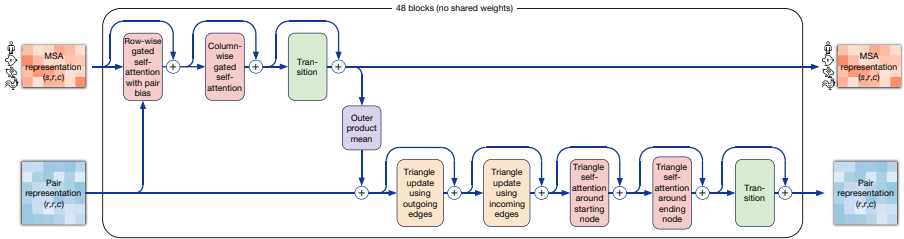
Fig. 3b, the pairs as directed edges

Maintains continuous spatial relationships before 3D coordinates are instantiated.

SI 1.5, Algorithm 4 (relative position), 1.7.1 (templates), 1.10 (recycling), 1.9.8 (distogram).

The Evoformer block

1. The sequences in an MSA are a set. The residues are a chain.



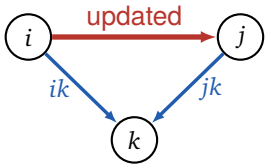
Axial attention alternates over sequence positions (rows) and homologous species (columns).

Bidirectional cross-talk: pairs bias MSA attention; outer product mean writes covariance into pairs.

[JEP+21] Jumper et al., *Nature* 2021, Fig. 3a (48 blocks, SI Algorithms 7–10).

Evoformer: triangle updates

2. Distances in 3D must close triangles.

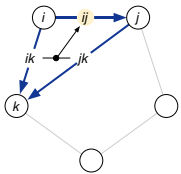


Edge (i, j) aggregates two-hop paths through intermediate nodes k: $\sum_k \mathbf{z}_{ik} \odot \mathbf{z}_{jk}$.

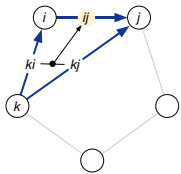
Enforces the 3D triangle inequality and metric consistency directly in 2D pair space.

Multiplicative updates

outgoing edges

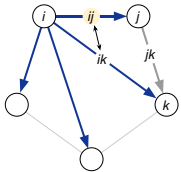


incoming edges

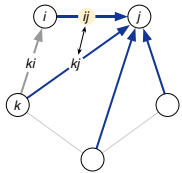


Triangle self-attention

starting node i



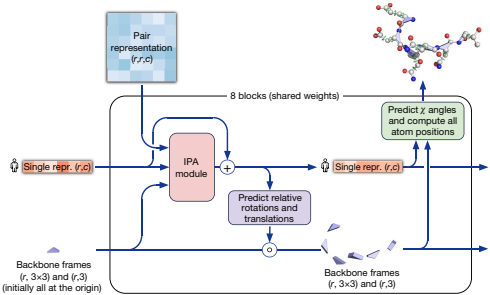
ending node j

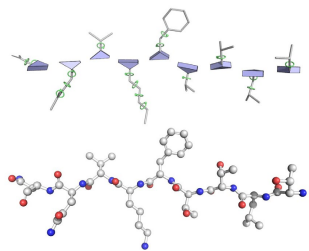


SI 1.6.4–1.6.6, Algorithms 11–14. Triangle multiplicative updates and axial attention.

Structure module: residues as rigid bodies

3. A structure is the same after any rotation or translation.





Residues are independent SE(3) rigid bodies $T_i = (R_i, \vec{\mathbf{t}}_i)$; dihedral angles χ place sidechains.

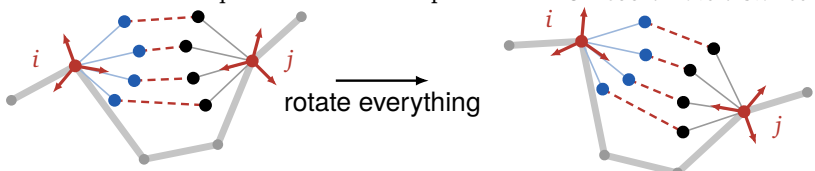
Simultaneous updates as an unconstrained residue gas; covalent chain geometry is learned, not hardcoded.

[JEP+21] Jumper et al., *Nature* 2021, Fig. 3c,d (8 iterations, SI Algorithms 20–23).

Invariant Point Attention (IPA)

3. A structure is the same after any rotation or translation.

$$a_{ij} = \mathrm{softmax}_j\Big(\underbrace{q_i^\top k_j}_{\text{scalar sequence attention}} + \underbrace{b_{ij}}_{\text{pair bias}} - \underbrace{\sum_p \left\| T_i \circ \vec{q}_{ip} - T_j \circ \vec{k}_{jp} \right\|^2}_{\text{3D coordinate distance penalty}} \Big)$$



Points are placed in the frames of i and j

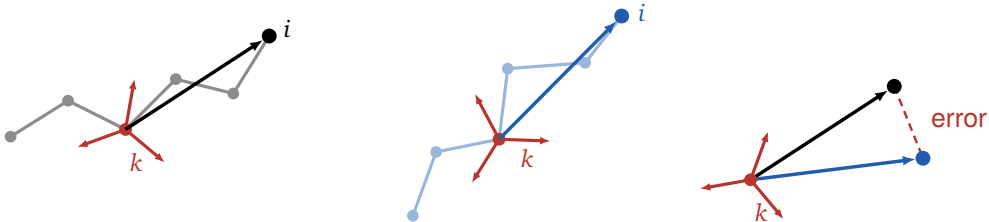
The distances stay the same

Exact algebraic $\mathrm{SE}(3)$-invariance: attention weights and scalar updates are independent of global frame.

SI 1.8.2, Algorithm 22 (single-head formulation). Algebraic invariance proof in backup 1.

Frame Aligned Point Error (FAPE)

4. A protein and its mirror image are different molecules.



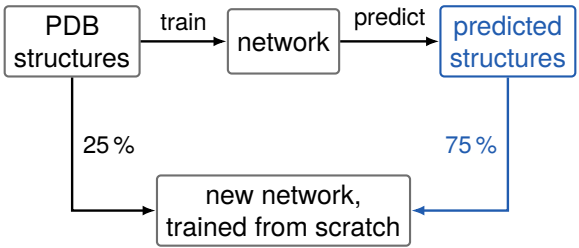
$$\text{FAPE} = \frac{1}{Z} \underset{k,i}{\text{mean}} \min\left(d_{\text{clamp}}, \left\| (\boldsymbol{R}_k, \boldsymbol{t}_k)^{-1} \boldsymbol{x}_i - (\boldsymbol{R}_k^{\text{true}}, \boldsymbol{t}_k^{\text{true}})^{-1} \boldsymbol{x}_i^{\text{true}} \right\|\right)$$

Local-frame point alignment: rigorously SE(3)-invariant, strictly sensitive to chiral enantiomer reflections.

Supplementary Information, Algorithm 28, with $Z = d_{\text{clamp}} = 10$ Å. Pseudometric proof in backup 2.

Training data: PDB and self-distillation

Self-distillation on 350k sequences triples training scale (75% batch share).



15% BERT masking trains representations directly on evolutionary sequences.

M ■ K V L I ■ G E L A F
M S K ■ L V N G E L A ■
M T R I ■ I N G D ■ A Y
■ T E V L I N ■ K L - F
M T ■ V L ■ N G K L A Y
M S D V L I N G ■ V A F

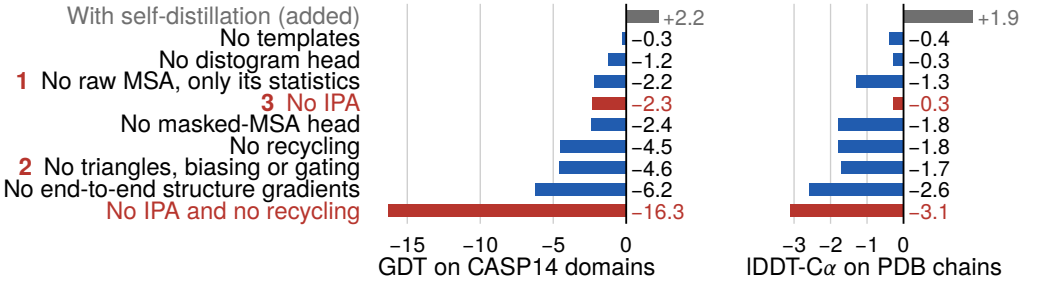
■ hidden and to be predicted

Dual self-supervision: leverages uncharacterized sequence databases to generalize beyond PDB coverage.

[XLHL20] Xie et al., CVPR 2020 (self-distillation). [DCLT19] Devlin et al., NAACL 2019 (BERT masking).

Ablations: the cost of each component

1 2 3 4

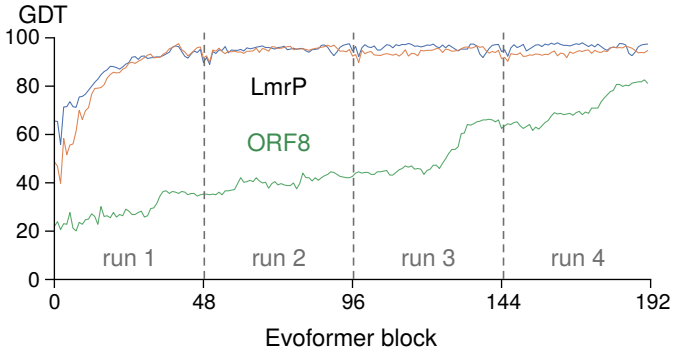


4 Without FAPE stereochemical grounding, reflection-invariant losses produce 50% inverted chiral enantiomers (Suppl. Fig. 9).

Severe non-linear coupling: removing both IPA and recycling causes a catastrophic −16.3 GDT collapse.

Redrawn from Fig. 4a, mean change vs baseline over training runs. GDT and lDDT 0–100. Row 3 (No IPA) removes iterative refinement but keeps frames and FAPE (SI 1.13.1). Line 4: Suppl. Fig. 9.

Intermediate structures in the trunk

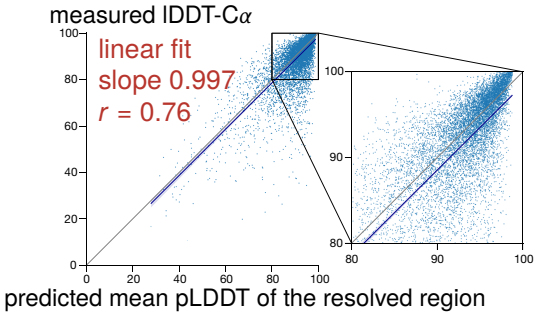


Shared Structure Module readouts probe intermediate coordinates across 192 unrolled blocks.

Bimodal folding dynamics: compact domains (LmrP) resolve early; complex topologies (ORF8) require full depth.

[JEP+21] Jumper et al., *Nature* 2021, Fig. 4b. LmrP: transporter (PDB 6T1Z); ORF8: SARS-CoV-2 (T1064).

Confidence: pLDDT against measured lDDT



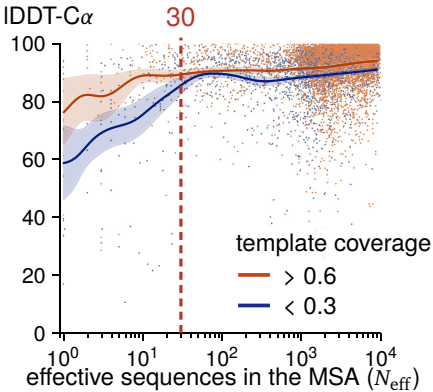
pLDDT faithfully predicts local lDDT-Cα (r = 0.76, slope 0.997 vs measured truth).

Scores below 50 reliably identify intrinsically disordered regions (IDRs).

Calibrated uncertainty: reliable residue-level error estimation without experimental ground truth.

[JEP+21] Jumper et al., *Nature* 2021, Fig. 2c, n = 10,795 chains. Following [TAW+21] Tunyasuvunakool et al., *Nature* 2021.

Limitations: MSA depth and complexes



Accuracy drops sharply below $N_{\text{eff}} \approx 30$ unless rescued by homologous templates.

Monomer models fail on obligate multimers stabilized by quaternary interfaces.

Predicts static, isolated ground states, omitting allosteric conformational ensembles.

[JEP+21] Jumper et al., *Nature* 2021, Fig. 5a.

After AlphaFold 2: 2021 to 2025

2021 RoseTTAFold published simultaneously [BDI+21]

98.5% of the human proteome predicted [TAW+21]

AlphaFold-Multimer: multimeric complexes [EOP+21]

2022 ESMFold: language model, single sequence [LAR+23]

2024 AlphaFold 3: all biomolecular complexes [AAD+24]

directly address the limits

Nobel Prize in Chemistry (Baker, Hassabis, Jumper, 2024)

2025 AlphaGenome: variant effects in non-coding DNA [Avs+25]

Expansion of structural genomics: from monomeric static predictions to proteome-scale databases and all-atom complexes.

Exception: AIchemy_RNA2 on CASP15 RNA targets. Complete citations on closing References slides.

AlphaFold 3: which choices survived

Abramson et al. [AAD+24], *Nature* 630 (2024), pages 493 to 498

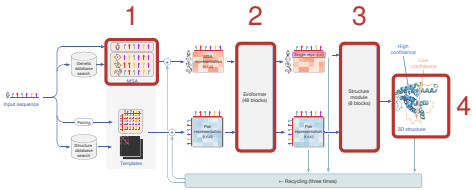
	AlphaFold 2	AlphaFold 3 verdict	The cost / patch
1	raw MSA attention	small MSA block, not kept	same MSA depth dependence
2	triangles on the pairs	kept unchanged, 48 blocks	inductive bias remains key
3	frames, IPA, equivariance	dropped for diffusion	clashes; ranking and distillation
4	FAPE in local frames	dropped; aligned MSE loss	4.4 % chirality errors; ranking, seeds

Unlike AlphaFold 2, AlphaFold 3 provides no component ablation study.

Generative paradigm shift: relational pair graphs survived; rigid kinematics yielded to all-atom diffusion.

[Sut19] R. Sutton, *The Bitter Lesson*, 2019. AF3 reports 4.4% chirality violations and relies on multi-seed ranking. Architecture: backup 5.

Summary: four design choices, measured



		GDT lost
1	attention over sequences	−2.2
2	triangles on the pairs	−4.6
3	frames and IPA	−2.3
4	FAPE in local frames	mirrored

Ablating any single structural inductive bias degrades accuracy; removing multiple causes catastrophic collapse.

Structural biology constraints as deep learning inductive biases: each choice quantitatively drives atomic accuracy.

[JEP+21] Jumper et al., *Nature* 2021. Figures reproduced under CC BY 4.0. CASP14: 0.96 Å median backbone error.

Exam Question

Machine Learning for Genomics (263-5351-00L)

Assigned Problem

Analyze architectural inductive biases, loss mechanics, and biological failure modes under shallow MSA alignments ($N_{\text{eff}} < 30$).

Core Dimensions

1. **Inductive bias**: Evoformer pair graph vs Invariant Point Attention
2. **Loss mechanics**: FAPE stereochemistry vs reflection invariance
3. **Genomics**: orphan proteins and quaternary complexes

First-principles theoretical analysis and empirical validation against CASP14 ablations.

Questions for the audience

1 AlphaFold 2 hardcoded SE(3) equivariance; AlphaFold 3 discarded it for diffusion. When does geometric inductive bias beat unconstrained generative modeling?

2 Accuracy collapses below $N_{\text{eff}} \approx 30$. Has the network learned folding biophysics, or evolutionary history? What experiment would decide this?

3 pLDDT is trained on self-prediction errors. Where does this confidence estimate fail, and how should pipelines interpret pLDDT < 50?

`github.com/BenBullinger/ML4Genomics-Presentation-AF2`

References: AlphaFold & Structural Biology

Primary literature and benchmark publications

[JEP+21] J. Jumper, R. Evans, A. Pritzel, et al., “Highly accurate protein structure prediction with AlphaFold,” *Nature* 596:583–589, 2021.

[TAW+21] K. Tunyasuvunakool, P. Adler, Z. Wu, et al., “Highly accurate protein structure prediction for the human proteome,” *Nature* 596:590–596, 2021.

[EOP+21] R. Evans, M. O’Neill, A. Pritzel, et al., “Protein complex prediction with AlphaFold-Multimer,” *bioRxiv*:2021.10.04.463034, 2021.

[BDI+21] M. Baek, F. DiMaio, I. Anishchenko, et al., “Accurate prediction of protein structures and interactions using a three-track neural network,” *Science* 373:871–876, 2021.

[AAD+24] J. Abramson, J. Adler, J. Dunger, et al., “Accurate structure prediction of biomolecular interactions with AlphaFold 3,” *Nature* 630:493–500, 2024.

[MHC+11] D. S. Marks, L. J. Colwell, R. Sheridan, et al., “Protein 3D structure computed from evolutionary sequence variation,” *PLoS ONE* 6:e28766, 2011.

References: Machine Learning & Genomics

Language models, diffusion, regulatory genomics, and paradigms

[LAR+23] Z. Lin, H. Akin, R. Rao, et al., "Evolutionary-scale prediction of atomic-level protein structure with a language model," *Science* 379:1123–1130, 2023.

[Avs+25] Ž. Avsec, et al., "Predicting variant regulatory impact in non-coding DNA with AlphaGenome," *bioRxiv*, 2025.

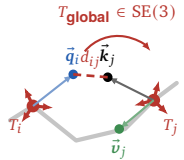
[DCLT19] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," *NAACL-HLT*, 2019.

[XLHL20] Q. Xie, M.-T. Luong, E. Hovy, Q. V. Le, "Self-training with noisy student improves ImageNet classification," *CVPR*, 2020.

[Sut19] R. Sutton, "The Bitter Lesson," *Incomplete Ideas* (essay), 2019.

Backup (1/5): Exact SE(3) invariance of IPA

SI 1.8.2, Algorithm 22, equations 3 to 6



→ **Frames** $T_i, T_j \in \text{SE}(3)$
• **Query** $\vec{q}_i \in \mathbb{R}^3$
• **Key** $\vec{k}_j \in \mathbb{R}^3$
• **Val** $\vec{v}_j \in \mathbb{R}^3$
−− **Dist.** d_{ij} invariant
T_{global} **Global** $\in \text{SE}(3)$

1. Attention Affinities: invariant under global rigid motion $T_{\text{global}} \in \text{SE}(3)$:

$$\left\| T_{\text{global}} \circ T_i \circ \vec{q}_i - T_{\text{global}} \circ T_j \circ \vec{k}_j \right\|^2 = \left\| T_{\text{global}} \circ (T_i \circ \vec{q}_i - T_j \circ \vec{k}_j) \right\|^2 = \left\| T_i \circ \vec{q}_i - T_j \circ \vec{k}_j \right\|^2$$

Rigid motion T_{global} is an isometry: preserves Euclidean norm identically $\implies$ a_{ij} is SE(3)-invariant.

2. Local Frame Projection: value aggregation mapped back to residue i frame:

$$(T_{\text{global}} \circ T_i)^{-1} \circ \sum_j a_{ij} \, (T_{\text{global}} \circ T_j) \circ \vec{v}_j$$

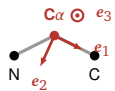
$$= T_i^{-1} \circ \underbrace{T_{\text{global}}^{-1} T_{\text{global}}}_{=\mathbf{I}} \circ \sum_j a_{ij} \, T_j \circ \vec{v}_j = T_i^{-1} \circ \sum_j a_{ij} \, T_j \circ \vec{v}_j$$

Group inverse $T_{\text{global}}^{-1} T_{\text{global}} = \mathbf{I}$ cancels algebraically $\implies$ updated coordinate representation is invariant.

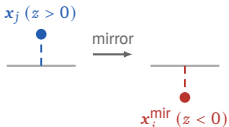
Exact algebraic SE(3)-invariance: attention weights and coordinate updates are independent of global pose.

Backup (2/5): Chirality and FAPE metric bounds

SI 1.8.1, 1.8.4, 1.9.4, Algorithm 28



Right-handed basis:
$\boldsymbol{e}_3 = \boldsymbol{e}_1 \times \boldsymbol{e}_2$



Reflection inverts z: $z \mapsto -z$

Gram–Schmidt local frames break reflection symmetry via z-axis inversion:

Mirror flip: $T_i^{-1} \circ \boldsymbol{x}_j \xrightarrow{\text{mirror}} \text{diag}(1, 1, -1)\,(T_i^{-1} \circ \boldsymbol{x}_j)$

FAPE loss: $\mathcal{L}_{\text{FAPE}}(X, X_{\text{mir}}) = \frac{2}{Z} \sum_{k,i} \min\left(d_{\text{clamp}}, \left|(T_k^{-1}\boldsymbol{x}_i)_z\right|\right) > 0$

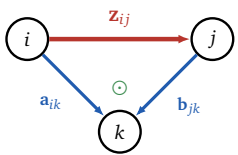
RMSD bound: $\mathcal{L}_{\text{FAPE}}(X, Y) \geq \min_{R \in \text{SO}(3),\, \boldsymbol{t}} \text{RMSD}(RX + \boldsymbol{t}, Y)$

Strictly positive for non-planar geometries ($z_i \neq 0$): detects chirality and bounds optimal RMSD without Kabsch SVD.

Breaks reflection symmetry without Kabsch alignment while bounding optimal RMSD.

Backup (3/5): Triangle updates and 2D metric consistency

SI 1.6.4–1.6.6, Algorithms 11–14



Metric closure: $d_{ij} \leq d_{ik} + d_{kj}$
Path marginalization: $\sum_k$ combines all k

1. Multiplicative Updates (SI Algorithm 11–12):

$$\mathbf{z}_{ij} \leftarrow \mathbf{z}_{ij} + \text{Linear}\left(\sum_{k=1}^{N} \sigma(\mathbf{a}_{ik}) \odot \sigma(\mathbf{b}_{jk})\right)$$

Gated Hadamard product $\odot$ of outgoing edges (i, k) and (j, k) completes triangular path (i, k, j).

2. Triangle Self-Attention (SI Algorithm 13–14):

$$\mathbf{A}_{ij} = \text{softmax}_j\left(\frac{1}{\sqrt{c}} \mathbf{q}_{ij}^{\top}\mathbf{k}_{ij} + \mathbf{b}_{ij} + \sum_{k=1}^{N} \mathbf{w}^{\top}\mathbf{z}_{ik}\right)$$

Edge (i, j) attends across the full graph, biased by third-edge marginalization $\sum_k \mathbf{w}^{\top}\mathbf{z}_{ik}$.

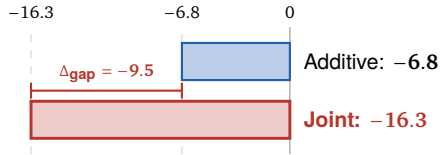
Enforces geometric consistency and metric embeddability directly in 2D pair space.

Backup (4/5): Non-linear ablation synergies and recycling

SI 1.10 and 1.13, Figure 4a

Ablated component	ΔGDT	Mechanism
No IPA alone	-2.3	Linear frame loss
No recycling alone	-4.5	Single-pass limit
No IPA + recycling	**−16.3**	**Collapse**
No raw MSA (DCA)	-2.2	Epistatic loss
No triangle updates	-4.6	Metric violation

Ablations reveal severe non-linear coupling: single errors compound without iterative correction.



$\Delta GDT(Joint) \ll \Delta GDT(IPA) + \Delta GDT(Rec)$

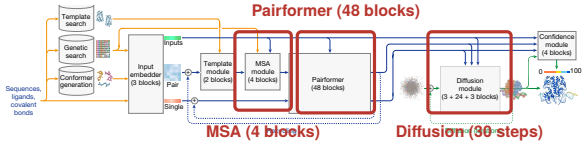
$-16.3 \ll -6.8$ (excess: -9.5 GDT)

4 cycles × 48 blocks = 192 effective Evoformer layers: recycling provides iterative depth for IPA.

Synergy: recycling provides the iterative trunk depth required for IPA to resolve complex topological knots.

Backup (5/5): AlphaFold 3: Diffusion vs Inductive Bias

Abramson et al. [AAD+24], *Nature* 630 (2024), Fig. 1d



Dimension	AlphaFold 2 (Inductive Bias)	AlphaFold 3 (Generative Diffusion)
Geometric space	Rigid residue gas $(R_i, \boldsymbol{t}_i) \in \mathrm{SE}(3)$	All-atom Cartesian coordinates $\boldsymbol{x} \in \mathbb{R}^{3N}$
3D structure module	Invariant Point Attention (IPA) + FAPE	Cartesian score-based diffusion model
Relational core	**Evoformer (48 triangle blocks)**	**Pairformer (48 triangle blocks retained!)**
Chirality guarantee	**Exact 0% violations** (Gram–Schmidt)	**4.4% violations** (requires 25 rollouts)

The relational pair graph survived; physical kinematic frames yielded to all-atom diffusion.